keylo

Key & BPM Detector — Technical & User Documentation

Version 1.0.0 · macOS (VST3/AU) · Windows (VST3) · Open Source

Keylo is a DAW plugin that detects the musical key and BPM of any audio running through it. Two independent pipelines — a hand-crafted DSP engine (TSA) and a fine-tuned ML model (S-KEY) — run in parallel and fuse their results. Everything runs locally. No internet connection required.

Repository	github.com/snowinch/keylo
License	Open Source — see LICENSE in repository
Support	typebeathouse@snowinch.com
Website	www.typebeathouse.com

Contents

Part I — User guide

1 Installation

macOS

Open the downloaded **.pkg** installer. Keylo installs both **VST3** and **AU** formats into the standard system plugin directories. After installation, restart your DAW and run a plugin scan — Keylo appears under the category **SNOW INCH**.

> ***Gatekeeper warning:*** *macOS may block the installer because the plugin is not yet Apple-notarised. To open it: right-click (or Control-click) the .pkg →* ***Open*** *→ confirm. Alternatively: System Settings → Privacy & Security → scroll to the blocked item →* ***Open Anyway****. It is safe to proceed.*

Windows

Run the **.exe** installer. Keylo installs as VST3 into *C:\Program Files\Common Files\VST3*. Restart your DAW and scan for new plugins — Keylo appears under **SNOW INCH**.

> ***SmartScreen warning:*** *Windows may show a 'Windows protected your PC' dialog because the installer is not yet code-signed. Click* ***More info*** *→* ***Run anyway****. It is safe to proceed.*

> *Keylo requires no internet connection after installation. The ML model is embedded inside the plugin binary (743 KB).*

2 Loading Keylo in your DAW

Load Keylo as an **insert effect** on any track. It does not modify the audio — it only reads and analyses it.

DAW	How to insert Keylo
FL Studio	Mixer → Insert FX slot → add Keylo (category: SNOW INCH)
Ableton Live	Audio Effects panel → drag onto track
Logic Pro	Channel strip → Audio FX → Keylo (AU, category: SNOW INCH)
Studio One	Insert FX chain → Keylo
Reaper	FX chain on any track → Add Keylo
Pro Tools	Inserts → Plug-In → Keylo (VST3)

For the most accurate reading, insert Keylo on the track with the main harmonic content (melody, chord loop, 808 channel). Loading on the master bus works too, but the full mix can reduce detection accuracy.

3 The interface

The plugin window is **380 × 310 px**, divided into five zones:

Zone	Height	Contents
Header	34 px	Mode pills (loop / melody) · Camelot toggle (cam) · Status dot
Hero	170 px	Large key display: root note + mode + Camelot code
BPM strip	40 px	BPM value (1 decimal) + confidence bar
Alt row	30 px	Two alternative keys + algorithm weight bar
Footer	36 px	Stability timeline · Reset button

The **algorithm weight bar** (right side of the alt row) shows which pipeline dominated the result: **teal = TSA**, **violet = S-KEY**.

Note notation

Keylo always uses **sharps only**: C, C#, D, D#, E, F, F#, G, G#, A, A#, B. Flats are never displayed.

Status indicator

Dot	Status	Meaning
●	**ready**	Detection complete — result is available
●	**listening**	Audio present — data is being collected
●	paused	No audio — plugin is idle

4 Controls and modes

Control	Function
loop / melody	Switches input mode (see below)
cam	Toggles between standard key notation and Camelot Wheel codes in the hero display
reset	Clears the audio buffer and restarts detection. Always press before switching to a different track or loop.

Input modes

Mode	Best for	Internal effect
loop	Full beats · sample packs · 808 tracks · chord loops	Weights bass analysis (HSBF, DTV, CTG) more heavily — better at finding the tonic from 808 patterns and kick placement
melody	Single-instrument leads · piano rolls · solo melodies	Weights the YIN pitch detector more heavily — more accurate when there is no strong bass reference

*When in doubt, use **loop** mode — the default for most trap and hip-hop scenarios.*

5 Workflow — step by step

1 **Insert Keylo on the track**
Add Keylo as an FX insert on the channel carrying the main harmonic content.

2 **Select input mode**
Choose **loop** for bass-heavy material, **melody** for single-instrument lines.

3 **Press Play**
Start playback. The status dot switches to teal ('listening') and collection begins immediately.

4 **Let the full track play through**
Keylo accumulates the audio buffer as the track plays. The result appears once the full audio has been processed. For looped material, at least one complete loop pass is required before detection stabilises.

5 **Read the key**
The hero display shows the detected key. Check the confidence bar and the stability timeline in the footer to assess reliability.

6 **Check alternatives if needed**
If the main result feels off, check the 'also:' row. On 808-heavy tracks the correct key often appears as the subdominant alternative.

7 **Press Reset before the next track**
Always clear the buffer before switching material. Leftover data will skew the next result.

6 Reading the result

The key display

Display	Meaning
C# minor	C-sharp natural minor
G Dorian	G Dorian — minor with a raised 6th, common in trap
A# minor	A-sharp minor (enharmonic B-flat minor)
D Phrygian	D Phrygian — dark mode with a lowered 2nd
F major	F major — less common in trap, more in R&B-influenced beats

Confidence

Confidence	Interpretation
80–100 %	Both pipelines agree — result is very likely correct
60–79 %	Pipelines mostly agree — reliable for most tracks
40–59 %	Some disagreement — check the alternative keys
< 40 %	Low certainty — track may be ambiguous, very short, or harmonically complex

Part II — Technical reference

7 Architecture overview

Keylo runs two fully independent key-detection pipelines in parallel and fuses their results. A DSP pipeline based on music-theory heuristics and an ML model trained on data capture complementary evidence. When they agree, confidence is boosted; when they disagree, the ML model is given higher weight.

The audio thread pushes samples into a ring buffer via the JUCE PluginProcessor. A background worker thread reads the buffer, runs both pipelines, and writes results to the UI thread via a lock-free queue. The audio thread is never blocked.

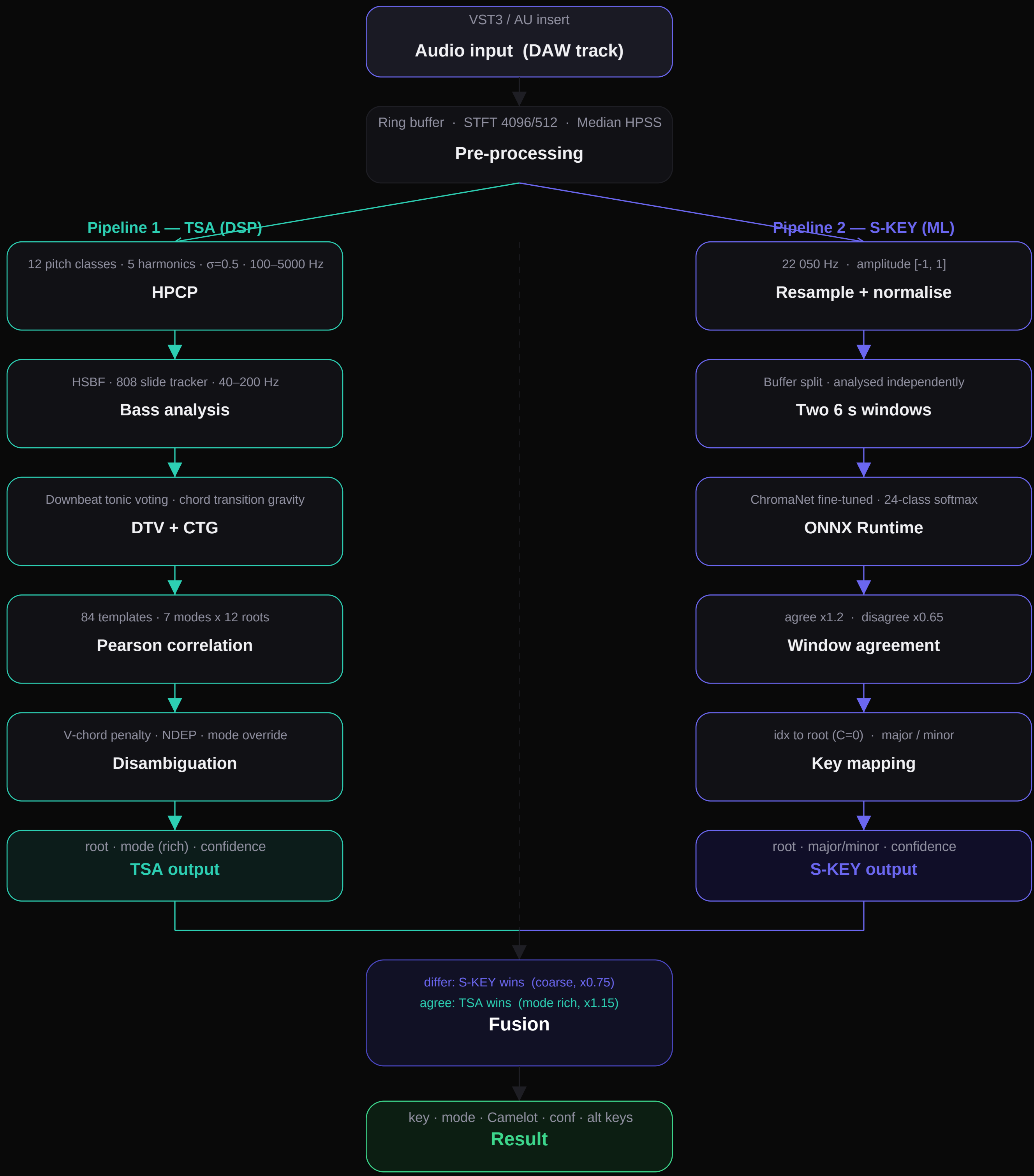

Figure 1 — Full detection pipeline. Teal = TSA path. Violet = S-KEY path.

8 Pipeline 1 — TSA (hand-crafted DSP)

TSA (Tonal Spectral Analysis) is a fully deterministic, rule-based pipeline written in C++17. It applies a sequence of music-theory-informed transforms to extract tonal evidence from the signal.

STFT and HPSS

The signal is framed with a 4096-point FFT and a 512-sample hop. A median HPSS filter (temporal window = 7 frames) suppresses drum transients before HPCP computation, preventing kick and snare energy from contaminating the chroma vector.

HPCP — Harmonic Pitch Class Profile

Collapses the magnitude spectrum into a 12-bin chroma vector (one bin per semitone).

Parameter	Value	Rationale
Frequency range	100–5000 Hz	Excludes sub-bass and very high partials
Harmonics	5	Sums the first 5 harmonic series members per candidate pitch
Gaussian spread	σ = 0.5 semitones	Smooths tuning deviations without blurring adjacent semitones
Onset weighting	up to ×4	Onsets contribute more — attacks carry more pitch information than sustain
Cadence weighting	×2 on last 25% of buffer	Cadential frames tend to land on the tonic

Bass analysis — HSBF and 808 slide tracker

HSBF (Harmonic Series Bass Fitting) scores each candidate bass fundamental by harmonic series fit, avoiding 3rd-harmonic contamination. Example: a C# bass at ~69 Hz has its 3rd harmonic at ~207 Hz (≈ G#) — without separation this would falsely inflate G# in the chroma.

808 slide tracker: at each bass onset, pitch is tracked for up to 15 frames post-attack. The first frame where pitch has stabilised (<8 cents change over 3 consecutive frames) is used as the note value. Handles the characteristic 808 pitch glide without picking up the transient.

DTV and CTG — tonic voting

DTV (Downbeat Tonic Voting): the beat with the highest bass-band onset novelty in each 4-beat bar votes for the tonic.

CTG (Chord Transition Gravity): scans for cadential resolution intervals (V→I = 7 st, bVII→I = 10, bVI→I = 8, bII→I = 1). Most resolutions received = tonic candidate.

Template correlation and disambiguation

Pearson correlation against 84 templates (7 modes × 12 roots, Electronic genre profile root=8.0, fifth=7.0). Three correction passes follow:

Pass	Purpose
V-chord disambiguation	Detects when the Pearson winner is 7 semitones above the DTV/CTG winner. Applies –0.40 penalty to the dominant and +0.60 boost to the tonic.
NDEP	Non-Diatonic Energy Penalty: corr –= 0.25 × foreignE / (diatonicE + foreignE). Can hurt Dorian tracks that use the raised 6th.
Mode override	Interval analysis overrides the mode label if a better fit is found.

9 Pipeline 2 — S-KEY (machine learning)

S-KEY is based on Deezer's **ChromaNet** architecture (VQT + CNN), with a custom fine-tuned variant **trained on 1 200 urban/trap tracks**, significantly improving accuracy on the harmonic patterns typical of contemporary trap and hip-hop. It outputs a softmax over **24 classes** (12 major + 12 minor roots).

Model

Property	Value
Architecture	VQT + CNN (ChromaNet, Deezer — fine-tuned)
Fine-tuning dataset	1 200 urban / trap tracks
Validation accuracy	71.75 %
Output classes	24 (12 major + 12 minor)
Model file	Assets/skey.onnx — 743 KB, embedded via BinaryData
Runtime	ONNX Runtime, bundled — no external dependency

Key mapping

S-KEY uses an A-relative index; mapping to Keylo's internal C=0 system:

```
Major (idx 0–11, A=0 … G#=11) root = (idx + 9) mod 12
Minor (idx 12–23, Bm=12 … Bbm=23) root = (idx - 12 + 11) mod 12
```

Two-window strategy

The ring buffer is split into two 6-second windows, each analysed independently. 6-second windows empirically outperform a single 12-second window.

Condition	Confidence adjustment
Both windows agree on root	Average confidence × 1.2
Windows disagree	Best window confidence × 0.65

ONNX export note (contributors)

Assets/skey.onnx is a single inline file — no external sidecar. ONNX Runtime cannot load models with external data from embedded binary. When re-exporting:

```
model = onnx.load('skey_finetuned.onnx', load_external_data=True)
onnx.save_model(model, 'skey_inline.onnx', save_as_external_data=False)
```

ml/models/skey_finetuned.onnx still has an external sidecar — do not embed it directly.

10 Fusion logic

Fusion runs in Analyser::runAnalysis() after both pipelines complete:

Condition	Root	Mode	Confidence	S-KEY weight
TSA root == S-KEY root	TSA root	TSA (rich: Dorian etc.)	max(tsa,skey) × 1.15	0.35
TSA root ≠ S-KEY root	S-KEY root	Natural minor or major	skey_conf × 0.75	0.85

When pipelines agree, TSA's richer mode label is preserved — S-KEY can only output major or minor; TSA can output Dorian, Phrygian, Lydian, Mixolydian, and other modes.

The algorithm weight bar reflects skeyWeight in real time: teal = 1 – skeyWeight (TSA share), violet = skeyWeight (S-KEY share).

11 Alternative keys

The 'also:' row always shows two alternatives:

Fusion outcome	Alt 1	Alt 2
S-KEY wins (skeyWeight ≥ 0.8)	TSA result	Subdominant of S-KEY result: (root + 5) mod 12, same mode
Pipelines agree (skeyWeight < 0.8)	Relative: minor → (root+3)%12 maj, major → (root+9)%12 min	Subdominant: (root + 5) mod 12, same mode

The subdominant is always present because the most common failure mode is locking onto the 5th instead of the root. Example: Keylo detects G#m → subdominant = C#m = real key.

12 BPM detection

BPM is detected in parallel using **aubio's onset-based tempo tracker**, independently from both TSA and S-KEY. Displayed to one decimal place with its own confidence bar.

The BPM engine analyses actual audio content — it does not read or lock to the DAW project tempo. Most reliable on material with clear transient pulses (kicks, hi-hats, claps). Unreliable on sustained pads with no rhythmic events.

13 Known limitations and future directions

Structural limitations that cannot be resolved by tuning parameters:

Limitation	Root cause	Impact
Fifth/root confusion on 808 tracks	Both Pearson and S-KEY analyse the full mix. When the 808 plays the 5th, both pipelines agree on the wrong root — fusion amplifies the error.	Main result 7 semitones off. Correct key often appears as subdominant alternative.
NDEP penalises Dorian	Dorian's raised 6th is non-diatonic to natural minor. NDEP subtracts from the correct key candidate.	Dorian tracks may be reported as natural minor.
Near-equal root/fifth weights	Electronic profile root=8.0, fifth=7.0. Dominant chord is a near-correct match for the tonic template.	Systematic dominant/tonic confusion in template correlation.
V-chord guard is overfit	Safety conditions calibrated to specific test tracks.	False positives/negatives on unseen material.
S-KEY has no mode granularity	24-class model: major/minor only.	Mode falls back to natural minor when pipelines disagree.
Small test set	9 loops. No major keys, no melody-only, no sparse bass.	Generalisation is limited.

Recommended future directions

1 — Source separation: low-pass at 250 Hz for bass, band-pass 250–5000 Hz for harmony. Highest-leverage architectural change.

2 — Replace Pearson with entropy minimisation: Gini coefficient or KL-divergence. Eliminates hand-crafted profile weights.

3 — Temporal phrase analysis: segment at spectral flux minima. Notes at phrase starts are strong tonic candidates.

4 — Expand test set: ≥ 50 loops covering major keys, sparse bass, melodic 808 lines, solo melodies, chromatic passing tones.

Appendix A — Build and install from source

Requirements: JUCE 8, CMake ≥ 3.22, C++17 compiler, ONNX Runtime (bundled as submodule).

```
git clone https://github.com/snowinch/keylo && cd keylo
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build --config Release
```

```
# Install to DAW (macOS)
cp -r build/Keylo_artefacts/Release/VST3/Keylo.vst3 ~/Library/Audio/Plug-Ins/VST3/
cp -r build/Keylo_artefacts/Release/AU/Keylo.component ~/Library/Audio/Plug-Ins/Components/
rm -rf ~/Library/Caches/AudioUnitCache # then restart DAW
```

Release builds are automated via GitHub Actions (.github/workflows/release.yml). Push a tag matching v* to trigger macOS (arm64) + Windows (x64) builds in parallel.

Appendix B — Test suite

```
./build/Release/KeyloTest Assets/test-loops/*.wav
```

Test loops are named Title - KEY - BPMbpm - artist.wav. The runner compares against both the main result and the alternative keys.

Track	Expected	Main result	Correct in alts?
Ice On Me	C#m	G#m (1A)	yes — alt 2 = C#m
Go Hard	Gm	Dm (7A)	yes — alt 2 = Gm
Dirty Sprite	Cm	Cm (5A)	yes — main
BNB	Dm	Dm (7A)	yes — main
Bape On Me	A#m	A#m (3A)	yes — main

3/5 exact on main result. 5/5 correct in main or alt.

Appendix C — Support

Bug reports and questions: **typebeathouse@snowinch.com** or Instagram DM **@typebeathouse**.

Website: **www.typebeathouse.com**